

Frequency-Optimized Local Region Common Spatial Pattern Approach for Motor Imagery Classification

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Abstract—This paper presents a novel feature extraction approach for motor imagery classification overcoming the weakness of conventional common spatial pattern (CSP) methods, especially for small sample settings. We consider local CSPs generated from individual channels and their neighbors (termed “local regions”) rather than a global CSP generated from all channels. The novelty is to select a few *good* local regions using interquartile range (IQR) or an “above the mean” rule based on variance ratio dispersion score (VRDS) and inter-class feature distance (ICFD); instead of computationally expensive cross-validation method. Furthermore, we develop frequency optimization using filter banks by extending the VRDS and ICFD to frequency-optimized local CSPs. The proposed methods are tested on three publicly available brain-computer interface (BCI) datasets: BCI competition III dataset IVa, BCI competition IV dataset I, and BCI competition IV dataset IIb. The proposed method exhibits substantially improved classification accuracy compared to recent related motor imagery (MI) classification methods.

Index Terms—Brain-computer interfaces (BCIs), electroencephalography (EEG), common spatial pattern (CSP), motor imagery (MI), local feature.

I. INTRODUCTION

BRAIN-COMPUTER INTERFACES (BCIs) enable communication between the human brain and external devices by converting brain signals into commands without the use of muscles or peripheral nerves [1]–[3]. Of the noninvasive brain signal measurement techniques magnetoencephalography (MEG), functional magnetic resonance imaging (fMRI) and electroencephalography (EEG), EEG has been the most studied and utilized because of its simplicity and good temporal resolution [4]. Event-related (de)synchronization (ERD/ERS) EEG patterns evident during motor imagery (MI) exhibit rhythmic component changes; these are important

features of the BCI system [5]. However, EEG signals suffer from non-stationary noise; the signal-to-noise ratio (SNR) is very low. Also, ERD/ERS patterns are highly dependent on individual characteristics in terms of event locations and frequency bands affected [6]. Therefore, many algorithms extracting ERD/ERS-related features have been developed in efforts to enhance the utility of EEG for BCI.

The common spatial pattern (CSP) is widely used to extract discriminative ERD/ERS-related features from EEG signals [7]–[10]. CSP solves the generalized eigenvalue problem posed by the sample covariance matrix obtained from EEG channel data, and determines the optimal spatial filters maximizing the output power for one class and minimizing it for other classes. The CSP approach requires that the sample covariance matrix be reliable, especially in a low-SNR environment [14]. Therefore, CSP performance deteriorates markedly in small-sample setting (SSS), where the sample covariance matrix can be very noisy because of the lack of adequate training samples [11].

To improve the reliability of covariance matrix estimations in SSS, a regularized common spatial pattern (R-CSP) approach [12] was developed to re-use the sample covariance matrices of other subjects with adjustment of diagonal terms [13], [14]. However, R-CSP performance is heavily dependent on the regularization parameters chosen. Optimization of R-CSP parameters is often very difficult, requiring computationally expensive cross-validation of parameters. The CSP patch (CSPP) approach described in [15] and [16] interprets the unreliability of a sample covariance in SSS as an over-fitting problem, and seeks to reduce the matrix size using several small sample covariance matrices obtained from subsets (‘patches’) of EEG channels. CSP is applied to the patches and the best three pairs of CSP filters from the patches in terms of the ‘ratio of medians’ are chosen to generate features; CSPP improves performance in the SSS environment.

On the other hand, the performance of CSP-based algorithms depends on the frequency range of EEG MI; performance is optimized when appropriate frequency bands are selected [17], [18]. Therefore, a frequency-optimized CSP algorithm, termed the filter-bank common spatial pattern (FBCSP), has been developed to compute CSP features for each frequency band (after passage through a filter bank); FBCSP selects best few frequency bands maximizing the mutual information between CSP features and their

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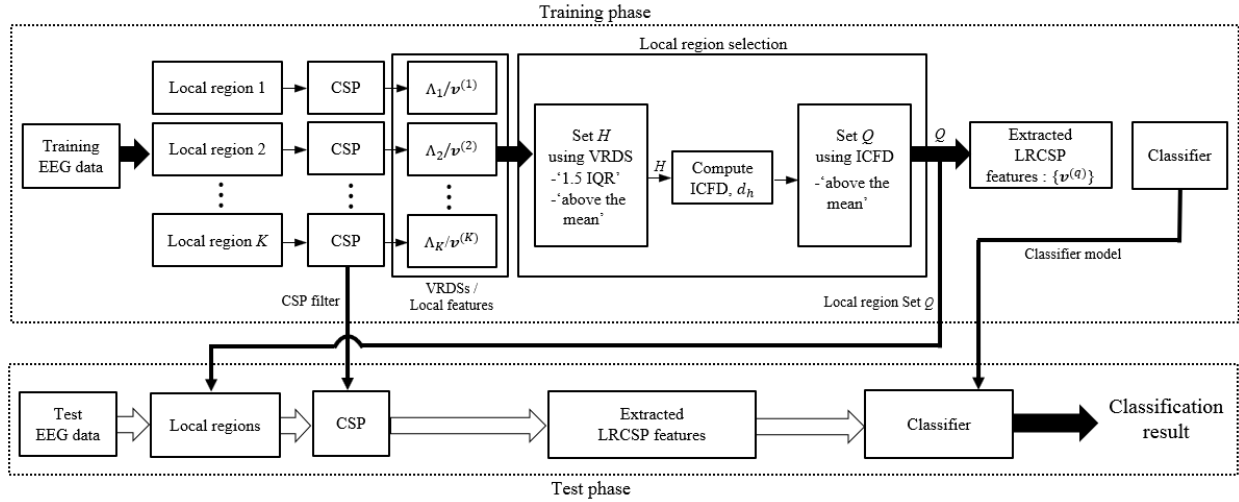


Fig. 1. Process of the local region common spatial pattern.

class label [19], [20]. This algorithm successfully extracts the optimized subject-related frequency bands and shows effective performance improvement.

Recently, several algorithms combining filter-bank structure with R-CSPs have been developed. The sub-band regularized common spatial pattern (SBRCS) uses principal component analysis (PCA) to extract R-CSP features from all frequency sub-bands [21]. The filter-bank regularized common spatial pattern (FBRCS) selects optimum frequency bands using mutual information of R-CSP features [22]. These algorithms, especially FBRCS, outperforms existing CSP-based BCI algorithms. However, they also suffer from the inherent drawbacks of R-CSPs, particularly the difficulties encountered when optimizing regularization parameters with cross-validation.

In this paper, we present a novel MI classification method for SSS based on CSP features extracted from EEG sub-channels (termed ‘local regions’ [23]) corresponding to all channels and optimized frequency bands. For each channel, we construct a subset of EEG channels (local region) with very adjacent channels and compute CSP features of the local region. We expect that small-size local sample covariance matrices would be robust in terms of noise and represent local brain activity at each channel more effectively [24]. Rather than CSPP, we select features from the local regions with large disparities in variance within their local CSP. We consider variance disparity to be a measure of the relevance of the region (or channel) to the MI task and the features from the *bad* regions which are irrelevant to the MI task as noise. The rationale behind selecting regions, not the filters as done in CSPP, is to eliminate the possibility that the filter selection process might be influenced by some *good* features generated by chance from the *bad* regions of which *badness* is determined by using all available data. The challenge posed by our local region common spatial pattern (LRCSP) approach is how to select CSP features of most contributed local regions for MI from the many such features evident in local regions.

First, we compute a variance ratio dispersion score (VRDS) (the difference between the largest and the smallest variance

ratio of a CSP filter) for each local region. For a large channel size case, we use 1.5 interquartile range (IQR) method [25] to select local regions corresponding to high outliers. For a small channel size, we select local regions for which the VRDSs lie above the mean VRDS; we term this the ‘above the mean’ rule. Second, using local regions chosen in this manner, we compute an inter-class feature distance (ICFD) and refine the local regions employing the ‘above the mean’ rule.

Furthermore, to optimize the frequency bands, we use FBCSP approach for local region CSPs. Using a filter bank structure, we select the two best frequency bands for each local region using mutual information. For frequency-optimized local regions that exhibit two frequency-band signals, we consider an extended VRDSs and apply the 1.5 IQR or the ‘above the mean’ rule depending on the channel size. Then, the extended ICFDs for frequency-optimized local regions are used to select more representative local regions based on the ‘above the mean’ rule. Thus, the selected local CSP features are now frequency-optimized; they are used as inputs of the support-vector machine (SVM) classifier [31]. The classification accuracy of the proposed algorithm is evaluated using three publicly available BCI datasets: BCI competition III dataset IVa, BCI competition IV dataset I, and BCI competition IV dataset IIb.

This paper is organized as follows. In section II, we formally introduce our method. In Section III, the experiments settings using the BCI competition III dataset IVa are explained. Section IV contains the experimental results. Finally, conclusion for this paper is made in Section V.

II. METHOD

The block diagram in Figure 1 illustrates our LRCSP algorithm. Let us consider EEG data of K measurement channels with L time samples. For each channel, we define a set of channels termed the local region; this consists of the individual channel and its adjacent immediate neighbor channels. We use the CSP to extract features of the local regions. During training stage, local regions that generate *good*

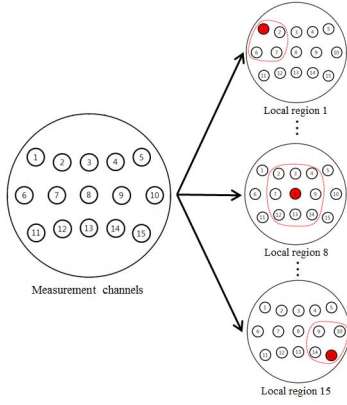


Fig. 2. Example of local region configuration. The dotted regions indicate local regions. The marked channels represent the main channel of each local region.

features (improving classification accuracy) are selected using two criteria, the VRDS and the ICFD. Features of *good* local regions are fed into the SVM classifier. This LRCSP approach is further generalized and optimized in the frequency domain using a filter bank.

A. Local Region and Local Region CSP

Consider a K -channel, spatial-temporal EEG data matrix $X \in \mathbb{R}^{K \times L}$, where L denotes the number of samples per channel. For each channel $k = 1, 2, \dots, K$, we define a set of channels, denoted by $R_k \subset \{1, 2, \dots, K\}$, consisting of main channel k and its immediate neighbor channels. The set R_k is termed a local sub-channel region of k , or simply the local region k . Let $|R_k|$ denote the number of channels in R_k , this may vary by the location of main channel k . Usually, $|R_k|$ ranges between 4 and 9, as the example in Figure 2 illustrates for $K = 15$. Note that the local regions are allowed to have common channels. Let $X^{(k)} \in \mathbb{R}^{|R_k| \times L}$ denote the matrix of EEG data for channels in the local region k . Assume that N trials of EEG training data are available, indexed $X_i \in \mathbb{R}^{K \times L}$, $i = 1, \dots, N$. Since we consider the binary MI classification, each trial belongs to one of the two index sets I_1 and I_2 , representing class 1 and 2 ($I_1 \cap I_2 = \emptyset$, $I_1 \cup I_2 = \{1, \dots, N\}$). Let $X_i^{(k)} \in \mathbb{R}^{|R_k| \times L}$ denote the i -th trial EEG training data of R_k . For each R_k , we consider a spatial filter ([7]) $\mathbf{p}^{(k)} \in \mathbb{R}^{|R_k|}$ to convert $|R_k|$ EEG signals in R_k into a single signal $\mathbf{p}^{(k)T} X_i^{(k)} \in \mathbb{R}^{1 \times L}$. The normalized variance of the signal $\mathbf{p}^{(k)T} X_i^{(k)}$ is given by:

$$\text{var}(\mathbf{p}^{(k)T} X_i^{(k)}) = \mathbf{p}^{(k)T} E_i^{(k)} \mathbf{p}^{(k)}, \quad (1)$$

where $E_i^{(k)} \in \mathbb{R}^{|R_k| \times |R_k|}$ is the normalized sample covariance matrix of $X_i^{(k)}$,

$$E_i^{(k)} = \frac{X_i^{(k)} X_i^{(k)T}}{\text{trace}(X_i^{(k)} X_i^{(k)T})}. \quad (2)$$

The averaged variance of the spatially filtered EEG signals from local region k for class c ($c \in \{1, 2\}$) can be written as

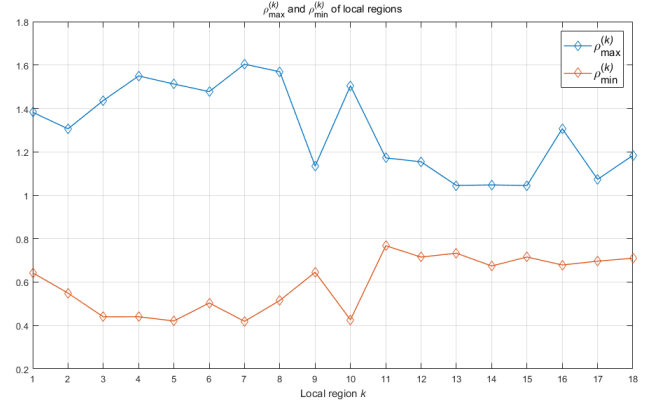


Fig. 3. Variance ratios for subject “ay” in BCI competition III dataset IVb.

$\mathbf{p}^{(k)T} \bar{E}^{(k,c)} \mathbf{p}^{(k)}$ using the averaged sample covariance matrix for the class c , denoted as $\bar{E}^{(k,c)}$:

$$\bar{E}^{(k,c)} = \frac{1}{|I_c|} \sum_{i \in I_c} E_i^{(k)}, \quad c \in \{1, 2\}, \quad (3)$$

where $|I_c|$ denotes the number of trials for the class c . Let $G(\mathbf{p}^{(k)})$ denote the average variance ratio between two classes, derived using a spatial filter $\mathbf{p}^{(k)}$:

$$G(\mathbf{p}^{(k)}) = \frac{\mathbf{p}^{(k)T} \bar{E}^{(k,1)} \mathbf{p}^{(k)}}{\mathbf{p}^{(k)T} \bar{E}^{(k,2)} \mathbf{p}^{(k)}}. \quad (4)$$

The CSP algorithm ([7]–[9]) finds the spatial filters maximizing or minimizing the averaged variance ratio G , and the corresponding variance ratio values as follows:

$$\begin{aligned} \mathbf{p}_{\max}^{(k)} &= \arg \max_{\mathbf{p}^{(k)}} G(\mathbf{p}^{(k)}), & \rho_{\max}^{(k)} &= \max_{\mathbf{p}^{(k)}} G(\mathbf{p}^{(k)}), \\ \mathbf{p}_{\min}^{(k)} &= \arg \min_{\mathbf{p}^{(k)}} G(\mathbf{p}^{(k)}), & \rho_{\min}^{(k)} &= \min_{\mathbf{p}^{(k)}} G(\mathbf{p}^{(k)}). \end{aligned} \quad (5)$$

Figure 3 shows sample plots of $\rho_{\max}^{(k)}$ and $\rho_{\min}^{(k)}$ relationships for the BCI competition datasets. For example, for the subject “ay” from BCI competition III dataset IVb, $\rho_{\max}^{(k)}$ and $\rho_{\min}^{(k)}$ are plotted for $k = 1, \dots, 18$. We consider the local regions with significant disparity among variance ratios as the regions that are highly relevant to the given MI task.

Finally, the local CSP feature vector for the i -th trial data in the local region k is defined as:

$$\mathbf{v}_i^{(k)} = \begin{bmatrix} v_{i,\max}^{(k)} & v_{i,\min}^{(k)} \end{bmatrix}^T, \quad (6)$$

where

$$\begin{aligned} v_{i,\max}^{(k)} &= \log \left(\text{var}(\mathbf{p}_{\max}^{(k)T} X_i^{(k)}) \right), \\ v_{i,\min}^{(k)} &= \log \left(\text{var}(\mathbf{p}_{\min}^{(k)T} X_i^{(k)}) \right). \end{aligned} \quad (7)$$

Compared to conventional CSP, where a global sample covariance matrix yields a feature vector, the proposed local region CSP approach generates the feature vectors from the sample covariance matrices of local regions. Of the K local regions, some may be more ‘meaningful’ than others and some may actually hinder MI classification. In fact, use of feature vectors

from all local regions generally does not entail the best classification performance, due either to the known high-dimensional feature problem [27] or possible noise from task-irrelevant regions. Therefore, we attempt to improve classification accuracy by using features from *good* (MI-relevant) regions with high-disparity variance ratios and discarding features from *bad* (MI-irrelevant) regions with low-disparity variance ratios.

B. Local Region Selection

To select *good* local regions, we use two criteria: the VRDS (the distance between the maximum and the minimum variance ratios), and the ICFD (the distance between the means of within-class features). Because CSP features are generated by CSP filters that maximize or minimize variance ratio of inter-class signals, it is reasonable to select regions wherein the difference between the maximum variance ratio, ρ_{max} and the minimum variance ratio, ρ_{min} are significant. Furthermore, among such local regions, we prefer the local regions that have large separation between the means of feature vectors within the same class. Although the two criteria seem to be closely related, successive screening using both criteria markedly improves classification performance.

1) **Variance Ratio Dispersion Score:** For a local region k , we define the difference between the maximum and the minimal variance ratio as the VRDS, denoted by Λ_k :

$$\Lambda_k := \rho_{max}^{(k)} - \rho_{min}^{(k)} \quad (8)$$

where $\rho_{max}^{(k)}$ and $\rho_{min}^{(k)}$ denotes the maximum and minimum variance ratio between classes obtained by p_{max} and p_{min} , respectively. Local regions with high VRDSs are expected to produce feature vectors affording high-level discrimination in terms of MI classification. Hence, we select local regions with VRDSs higher than a threshold, denoted by Λ_{thr} , defined using one of two strategies depending on channel size. For a large channel size K , we use an IQR method originally employed to identify outliers [25]:

$$\Lambda_{thr} := q_3 + 1.5(q_3 - q_1), \quad (9)$$

where q_1 is the value 25%-percentile of VRDSs, and q_3 is the value 75%-percentile of VRDSs. When the Λ_{thr} is larger than $\max\{\Lambda_k\}$, it means the threshold using IQR is too high. In such cases, which often occurs for a small K , the threshold is set to the average of all VRDS to select more meaningful local regions:

$$\Lambda_{thr} := \frac{1}{K} \sum_{k=1}^K \Lambda_k. \quad (10)$$

Let H denote the index set of local regions, the VRDSs of which are greater than Λ_{thr} :

$$H = \{k | \Lambda_k > \Lambda_{thr}\}. \quad (11)$$

Note that the threshold Λ_{thr} and the H are subject-dependent, and H is always not empty, for $\arg \max_k \{\Lambda_k\} \in H$ always.

2) **Inter-Class Feature Distance:** The mean feature vector for a class c in a local region h , denoted by $\bar{\mathbf{v}}^{(h,c)}$ is given by:

$$\bar{\mathbf{v}}^{(h,c)} = \frac{1}{|I_c|} \sum_{i \in I_c} \mathbf{v}_i^{(h)}. \quad (12)$$

Then, the ICFD is:

$$d_h = \|\bar{\mathbf{v}}^{(h,1)} - \bar{\mathbf{v}}^{(h,2)}\|, \quad (13)$$

where $\|\cdot\|$ denotes the ℓ_2 norm. Finally, we define our desired index set of *good* local regions, termed a feature region set denoted by Q , by choosing local regions exhibiting higher ICFDs than the average:

$$Q = \left\{ h \in H \mid d_h > \frac{1}{|H|} \sum_{h \in H} d_h \right\}. \quad (14)$$

Again, Q is not empty, since $\arg \max_h \{d_h\} \in Q$.

C. Local Region CSP Features and Classifier Training

Using training data, all local CSP filters and the feature region set Q are determined. For the trial data $X_1, \dots, X_N$, LRCSP feature set V is given by:

$$V = \left\{ \mathbf{v}_i^{(q)} \mid i = 1, \dots, N, q \in Q \right\}. \quad (15)$$

In training mode, V values and their associated class label vector $\mathbf{c} = [c_1, c_2, \dots, c_N]^T \in \mathbb{R}^N$ are used to train the SVM classifier. Algorithm 1 summarizes the process of the proposed method.

Algorithm 1 The Framework of LRCSP Algorithm

Input: Binary MI EEG data for each class of subject, where class label $c = \{1, 2\}$.

Output: LRCSP feature set.

for $u \leftarrow 1$, the number of subjects **do**

for $k \leftarrow 1$, the number of EEG channels **do**

 · Configure local region k and extract local region k EEG data $X^{(k)}$.

 · Construct the average normalized sample covariance matrix of class c , $\bar{E}^{(k,c)}$ using Eq. (3).

 · Extract the $p_{max}^{(k)}$, $p_{min}^{(k)}$ and $\rho_{max}^{(k)}$, $\rho_{min}^{(k)}$ following Eq. (5).

 · Extract the local CSP feature vector using Eq. (6).

 · Calculate the VRDS and threshold using Eq. (8), (9), (10).

end for

 · Set H using VRDS following Eq. (11).

 · Calculate the ICFD of local region $h \in H$ using Eq. (13).

 · Set Q using ICDF following Eq. (14).

 · **Output** LRCSP feature set following Eq. (15).

end for

Note that both our algorithm and the CSPP ([15], [16]), employ several small-sized sample covariance matrices rather than global matrices. The two algorithms differ in two respects. The CSPP performs the selection on the filter level, but LRCSP performs on the local region (or channel) level. Also, the CSPP

selects a fixed number of *good* features (for example 6 in [15]) from the patches; our algorithm selects local regions that exceed thresholds that are systematically defined using two criteria. The performance comparison of Section IV confirms that our algorithm is superior to the CSPP.

D. Frequency Domain Optimized Local Region CSP

CSP performance is dependent on the frequency range of EEG MI data. Here, we develop a frequency-optimized local region CSP algorithm using a filter bank; we term this a local region frequency-optimized CSP (LRFCSPP). Using a filter bank with multiple, say J , passband filters, J datasets in different frequency bands are obtained for each local region. For each local region, the best two frequency bands, which may vary with the local region, are selected based on the mutual information criterion of the filter-bank CSP algorithm [20], [26]. The feature vector representing a frequency-optimized local region is defined as a four dimensional vector by concatenating the local CSP feature vectors for two frequency bands. For LRFCSPP, the *good* local regions are selected similarly to the LRCSP proposed in the previous subsection using a modified VRDS criterion and ICFD extended for four dimensional feature vectors.

1) *Filter Bank Selection for a Local Region*: Consider a filter bank with J passband filters producing J sub-band signals. Let $X^{(k,j)} \in \mathbb{R}^{|R_k| \times L}$ denote the matrix of EEG data containing the j -th frequency band EEG signals from local region k . Let $\mathbf{p}^{(k,j)} \in \mathbb{R}^{|R_k|}$ denote the spatial filter used to process EEG signals for the local region k in the j -th frequency band. Using the CSP, we select two spatial filters that maximizes and minimizes the variance ratio, $\mathbf{p}_{\max}^{(k,j)}$ and $\mathbf{p}_{\min}^{(k,j)}$, respectively, and their corresponding variance ratio, $\rho_{\max}^{(k,j)}$ and $\rho_{\min}^{(k,j)}$. The features for local region k in the j -th frequency band at i -th trial are:

$$\begin{aligned} v_{i,\max}^{(k,j)} &= \log \left(\text{var}(\mathbf{p}_{\max}^{(k,j)T} X_i^{(k,j)}) \right), \\ v_{i,\min}^{(k,j)} &= \log \left(\text{var}(\mathbf{p}_{\min}^{(k,j)T} X_i^{(k,j)}) \right), \\ i &= 1, 2, \dots, N, \quad j = 1, 2, \dots, J. \end{aligned} \quad (16)$$

To select optimum frequency bands for the local region k , the mutual information $I(\{v_{i,\max}^{(k,j)}\}; c)$ and $I(\{v_{i,\min}^{(k,j)}\}; c)$:

$$\begin{aligned} I(\{v_{i,\max}^{(k,j)}\}; c) &= H(c) - \sum_{c=1}^2 \sum_{i=1}^N p(c|v_{i,\max}^{(k,j)}) \log_2 p(c|v_{i,\max}^{(k,j)}), \\ I(\{v_{i,\min}^{(k,j)}\}; c) &= H(c) - \sum_{c=1}^2 \sum_{i=1}^N p(c|v_{i,\min}^{(k,j)}) \log_2 p(c|v_{i,\min}^{(k,j)}), \end{aligned} \quad (17)$$

are computed based on the estimated $p(v_{i,\max}^{(k,j)}|c)$ and $p(v_{i,\min}^{(k,j)}|c)$ using Parzen Window [20]. Then, the two distinct frequency bands that have highest mutual information are selected for local region k . Let $J_1^{(k)}$ and $J_2^{(k)}$ denote the selected two frequency bands for local region $k = 1, 2, \dots, K$. The frequency-optimized local CSP feature vector is defined

as a four dimensional vector as the following:

$$\mathbf{w}_i^{(k)} = \begin{bmatrix} v_{i,\max}^{(k,J_1^{(k)})} & v_{i,\min}^{(k,J_1^{(k)})} & v_{i,\max}^{(k,J_2^{(k)})} & v_{i,\min}^{(k,J_2^{(k)})} \end{bmatrix}^T. \quad (18)$$

2) *Frequency-Optimized Local Region Selection*: We now have two pairs of variance ratios $\rho_{\max}^{(k,J_1^{(k)})}$, $\rho_{\min}^{(k,J_1^{(k)})}$ for the $J_1^{(k)}$ frequency band and $\rho_{\max}^{(k,J_2^{(k)})}$, $\rho_{\min}^{(k,J_2^{(k)})}$ for the $J_2^{(k)}$ frequency band. We define the VRDS of the frequency-optimized local region k , denoted as Λ_k^{freq} , as the mean of the VRDSs in each frequency band:

$$\Lambda_k^{freq} := \frac{1}{2} \left(\rho_{\max}^{(k,J_1^{(k)})} - \rho_{\min}^{(k,J_1^{(k)})} + \rho_{\max}^{(k,J_2^{(k)})} - \rho_{\min}^{(k,J_2^{(k)})} \right). \quad (19)$$

For a large channel case, a threshold Λ_{thr}^{freq} is set to based on 1.5 IQR rule:

$$\Lambda_{thr}^{freq} := q_3^{freq} + 1.5(q_3^{freq} - q_1^{freq}), \quad (20)$$

where q_1^{freq} is the middle value between the minimum and median of the VRDSs and q_3^{freq} is the middle value between the median and maximum of the VRDSs. If a small channel case or the Λ_{thr}^{freq} is larger than $\max \{\Lambda_k^{freq}\}$, we use the ‘above the mean’ rule:

$$\Lambda_{thr}^{freq} := \frac{1}{K} \sum_{k=1}^K \Lambda_k^{freq}, \quad (21)$$

Then, the frequency-optimized discriminative local region set H_{freq} (based on Λ_k^{freq}) is as follows:

$$H_{freq} = \{k \in \{1, 2, \dots, K\} | \Lambda_k^{freq} > \Lambda_{thr}^{freq}\}. \quad (22)$$

H_{freq} is always not empty, for $\arg \max_k \{\Lambda_k^{freq}\} \in H_{freq}$ always. Using the mean frequency-optimized local CSP feature for a class c :

$$\bar{\mathbf{w}}^{(h,c)} = \frac{1}{|I_c|} \sum_{i \in I_c} \mathbf{w}_i^{(h)}, \quad (23)$$

we compute the frequency-optimized ICFD of the local region h :

$$d_h^{freq} = \|\bar{\mathbf{w}}^{(h,1)} - \bar{\mathbf{w}}^{(h,2)}\|. \quad (24)$$

The feature region set Q^{freq} for the LRFCSPP is the set of local regions with d_h^{freq} -values higher than the average:

$$Q_{freq} = \left\{ h \in H_{freq} | d_h^{freq} > \frac{1}{|H_{freq}|} \sum_{h \in H_{freq}} d_h^{freq} \right\}. \quad (25)$$

Also, Q_{freq} is not empty, since $\arg \max_h \{d_h^{freq}\} \in Q_{freq}$.

Again, with training data, Q_{freq} , $J_1^{(q)}$, $J_2^{(q)}$, $\mathbf{p}_{\max}^{(q,J_1^{(q)})}$ and $\mathbf{p}_{\min}^{(q,J_2^{(q)})}$ are all determined. For the data set $X_1, \dots, X_N$, the LRFCSPP feature set are given as:

$$V_{freq} = \{\mathbf{w}_i^{(q)} | i = 1, \dots, N, q \in Q_{freq}\}. \quad (26)$$

In the training mode, V_{freq} and their class label vector $\mathbf{c}$ are fed to the SVM classifier. Figure 4 shows a block diagram of our LRFCSPP.

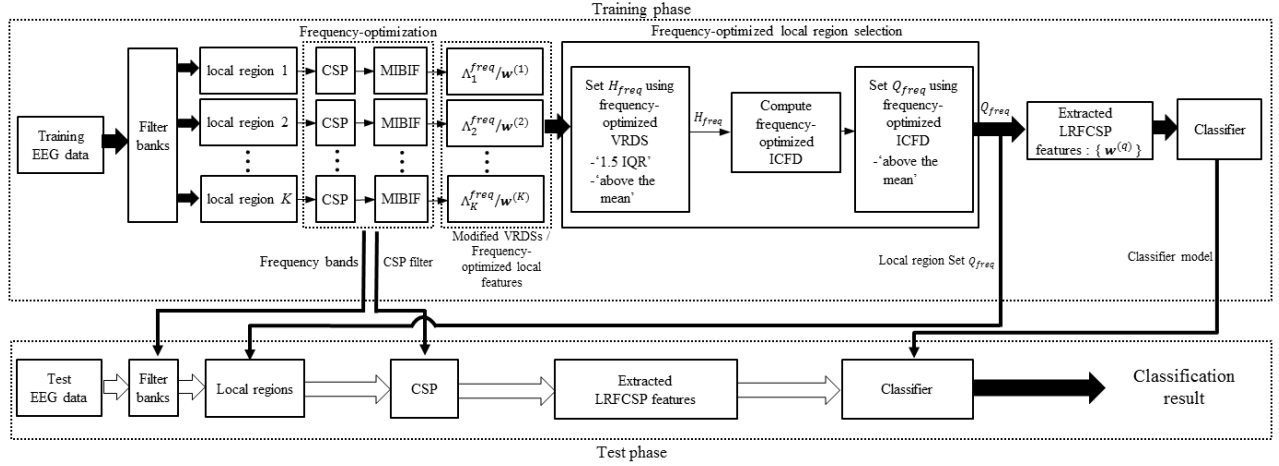


Fig. 4. The block diagram of LRFCSP.

TABLE I
BCI COMPETITION III DATASET IVA

Subject	Training data	Test data
al	224	56
aa	168	112
aw	84	196
av	56	224
ay	28	252

III. DATA AND EXPERIMENTS

A. Data Description

To evaluate the performance of our proposed method, we use the publicly available dataset IVa of BCI Competition III [28], dataset I of BCI competition IV [1], and dataset IIb of BCI competition IV. The BCI competition III dataset IVa is useful to evaluate the performance with small training sets and has been used to test many variants of conventional CSP algorithms [29]. EEG data were obtained from five healthy subjects ("al", "aa", "av", "aw" and "ay") who participated in a motor imagery experiment. Each subject was asked to imagine the movement of the right hand or foot after a visual cue was delivered. EEG data were recorded using a 118-channel electrodes ($K = 118$), bandpass-filtered between 0.05Hz and 200 Hz, and sampled at 100Hz. All subjects performed 280 trials, 140 per class, divided into training and test datasets. The number of trials for the training data and the test data of the five subjects are shown in Table I.

The BCI competition IV dataset I includes data recorded from seven different subjects and contains two MI EEG signal classes. 59 channels ($K = 59$) of EEG data were recorded at a sampling rate of 1000Hz and bandpass-filtered between 0.05Hz and 200Hz. The down sampled signal at 100 Hz has been used and each subject performed 200 trials, i.e., 100 trials per classes.

The BCI competition IV dataset IIb was recorded from nine subjects and three channels (C3, Cz, and C4) at a sampling rate of 250Hz for left hand and right hand MI tasks. Data were bandpass-filtered between 0.5 and 100Hz. As in [30],

TABLE II
LOCAL REGION CONFIGURATION FOR 18 CHANNELS ($K = 18$)

Number	Channels in local region
1	R_{T7} : T7 , CFC5, C3, CCP5
2	R_{CFC5} : CFC5 , T7, C3, CFC3
3	R_{CFC3} : CFC3 , CFC5, C3, C1, Cz
4	R_{C3} : C3 , T7, CFC5, CFC3, CP5, CCP3, C1
5	R_{C1} : C1 , CFC3, C3, CCP3, Cz
6	R_{CCP5} : CCP5 , T7, C3, CCP3, P3
7	R_{CCP3} : CCP3 , C3, C1, CCP5, P3, Cz, Pz
8	R_{P3} : P3 , CCP5, CCP3, Pz
9	R_{Cz} : Cz , C1, Pz, C2
10	R_{Pz} : Pz , CCP3, P3, Cz, CCP4, P4
11	R_{C2} : C2 , Cz, CFC4, C4, CCP4
12	R_{CFC4} : CFC4 , Cz, C2, CFC6, C4
13	R_{CFC6} : CFC6 , CFC4, C4, T8
14	R_{C4} : C4 , C2, CFC4, CFC6, T8, CCP4, CCP6
15	R_{T8} : T8 , CFC6, C4, CCP6
16	R_{CCP4} : CCP4 , Cz, C2, C4, CCP6, P4, Pz
17	R_{CCP6} : CCP6 , C4, T8, CCP4, P4
18	R_{P4} : P4 , Pz, CCP4, CCP6

only the third session data were used for evaluation. Each subject performed 160 trials, i.e., 80 trials per classes.

B. Data Processing

For the experiments, we use EEG data derived from 0.5s to 3s after the visual cue. The data are bandpass-filtered using a fourth-order Butterworth filter operating at 7-30Hz for LRCSP. For LRFCSP, eight filter banks are used dividing the frequency range from 4 to 36Hz evenly with 4Hz.

In BCI competition III dataset IVa, not all 118 channels are often used in MI classification studies. For example, CSPP in [15] uses 59 channels ($K = 59$) resulting from 15 patches each of those has 12 patch member channels around a center channel as depicted in Figure 5. The R-CSP ([14]), FBCSP ([19]), SBRCSP ([21]), and FBRCSP ([22]) use 18 channels ($K = 18$) selected using homunculus theory [32] as drawn in Figure 6. The local region configuration of the 18 channels are listed in Table II; the channel in bold indicates the main channel of local region. Here, we evaluate scenarios in which K is either 59 or 18. For $K = 59$, we use the 1.5 IQR

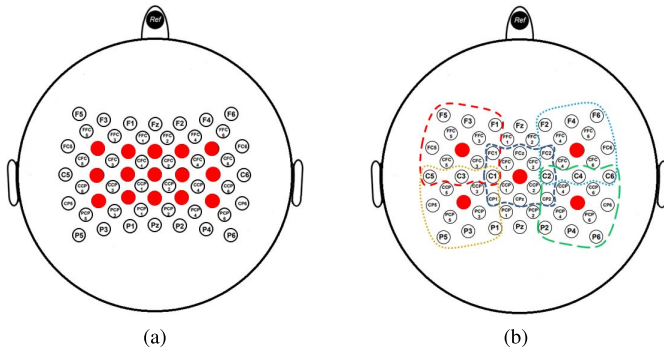


Fig. 5. Location of 59 channels ($K = 59$) for BCI competition III dataset IVa and patch configuration of CSPP. (a) Location of 59 channels and 15 channels marked in red indicate the center channels of patches. (b) Example of patch configuration of 5 center channels.

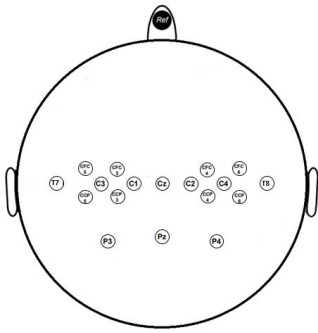


Fig. 6. Locations of the used 18 channels ($K = 18$) for BCI competition III dataset IVa.

method to define Λ_{thr} ; for $K = 18$ we use the ‘above the mean’ approach to derive Λ_{thr} .

For BCI competition IV dataset I, we use 59 channels ($K = 59$) to evaluate the performance. We apply the 1.5 IQR method to define Λ_{thr} .

In BCI competition IV dataset IIb, three local regions are constructed to evaluate the performance, i.e., local region 1: C3-CZ, local region 2: C3-Cz-C4, and local region 3: Cz-C4. We apply the ‘above the mean’ to define Λ_{thr} . Since only three local regions are configurable, we extract feature set using only VRDS criterion.

IV. RESULT AND DISCUSSION

In the section, we present the experimental results for the three aforementioned BCI competition datasets. First, we apply the LRCSP to 59 channels from BCI competition III dataset IVa and evaluate its performance compared to CSPP. Second, we compare the performance of LRCSP with R-CSP on 18 channels from BCI competition III dataset IVa. Third, using 18-channel data from BCI competition III dataset IVa, we evaluate the performance of LRCSP compared to frequency-optimized CSP methods, including FBCSP, SBRCSP and FBRCSP. Finally, we present the cross-validation results of proposed methods in comparison with the recent related methods, including CSP-tangent space mapping (CSP-TSM) ([33]), sparse Bayesian learning filter

TABLE III
CLASSIFICATION ACCURACY OF LRCSP AND CSPP FOR 59 CHANNELS ($K = 59$) BCI COMPETITION III DATASET IVa

subject	CSPP	LRCSP
al	98.21	98.21 (13)
aa	77.68	79.46 (13)
av	71.94	72.45 (2)
aw	84.82	87.95 (14)
ay	87.3	87.7 (1)
mean	83.99	85.15

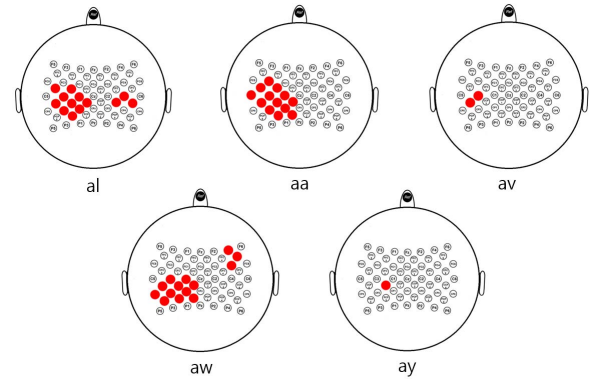


Fig. 7. Main channel location of selected local regions by LRCSP for five subject.

bank (SBLFB) ([34]), temporal filter parameter optimization CSP (TFPO-CSP) ([35]), and improved discriminative FBCSP (I-DFBCSP) ([36]) using BCI competition III dataset IVa, BCI competition IV dataset I, and dataset IIb.

A. Performance Comparison of LRCSP With CSPP

To compare the performance of LRCSP and CSPP, we use the 59 channels ($K = 59$) employed for the CSPP in [15] and 1.5 IQR method to calculate Λ_{thr} . Table III presents the classification accuracies of our LRCSP and the CSPP for the BCI competition III dataset IVa. The numbers in parentheses are those of selected local regions. Figure 7 shows the location of the main channels for each local region selected by LRCSP for five subjects. The number of selected local regions for subjects “al”, “aa”, “av”, “aw”, and “ay” are 13, 13, 2, 14, and 1, respectively. Compared to the CSPP, our LRCSP shows a higher classification accuracy, except for subject “al”. The mean classification accuracy is 1.16% better than that of the CSPP. Unlike the CSPP, which extracts a fixed number of features, the LRCSP extracts a subject-dependent number of features from selected local regions, thus improving classification accuracy.

B. Performance Comparison of LRCSP With R-CSP

In this experiment, we present the experimental performance result of our LRCSP method in comparison with the R-CSP. Table IV shows the classification accuracies of the test datasets for 18 channels ($K = 18$) of BCI competition III dataset IVa. The numbers in parentheses are those of selected local regions. Compared to the R-CSP, the LRCSP improves

TABLE IV

CLASSIFICATION ACCURACY OF LRCSP AND R-CSP FOR 18 CHANNELS ($K = 18$) BCI COMPETITION III DATASET IVA

subject	R-CSP	LRCSP
al	91.07	100 (4)
aa	75.89	77.23 (4)
av	60.2	70.92 (4)
aw	70.98	85.71 (3)
ay	88	90.48 (5)
mean	77.23	84.87

TABLE V

CLASSIFICATION ACCURACY OF LRFCSP AND FREQUENCY OPTIMIZED CSP METHODS FOR 18 CHANNELS ($K = 18$) BCI COMPETITION III DATASET IVA

subject	FBCSP	SBRCSP	FBRCS	LRFCSP
al	94.64	98.21	94.64	100 (3)
aa	88.39	86.61	91.07	89.29(5)
av	61.22	63.78	75	73.47(4)
aw	80.02	89.05	76.78	89.73 (6)
ay	82.14	77.78	93.65	94.05 (4)
mean	81.28	83.09	86.23	89.31

classification for every subject. The mean classification accuracy is 7.64% better than afforded by the R-CSP. This result shows that the LRCSP can improve performance by directly selecting meaningful local regions in the absence of cross-validation, unlike the R-CSP, where performance is dependent on cross-validation for parameter selection.

C. Performance Comparison of LRFCSP With Frequency-Optimized CSP Based Methods

Table V lists the classification accuracies of the test datasets afford by our LRFCSP and frequency-optimized CSP based methods for 18 channels ($K = 18$) of BCI competition III dataset IVa. The numbers of local regions selected are shown in parentheses. The LRFCSP yields better classification accuracies for subject “al”, “av” and “ay” than frequency-optimized conventional CSP-based methods. The mean LRFCSP classification accuracy is better than those of other methods, being 8.03%, 6.22%, and 3.08% higher than those of the FBCSP, SBRCSP, and FBRCS, respectively. Figure 8 shows the selected local regions of LRFCSP for subject “al”, “aa”, “av”, “aw” and “ay”; there are numbered 3, 5, 4, 6, and 4 respectively. Local region R_{C3} (C3,T7,CFC5,CFC3,CCP5,CCP3,C1) (red dotted line in Figure 8) and R_{CCP3} (CCP3,C3,C1,CCP5,P3,Cz,Pz) (dark blue dotted line in Figure 8) are selected for every subjects. As shown in Figure 8, the numbers and locations of selected local regions differ for each subject. The FBRCS and SBRCSP performed well for subjects “av”, “aw”, and “ay” with mean classification accuracies of 81.81% and 76.87%, respectively. This results show the effect of R-CSP based method in SSS environment. However, the mean LRFCSP classification accuracy for three subject is 85.75%, better than the FBRCS and SBRCSP. Especially, the LRFCSP entails the highest accuracy for subject “ay”, who contributed the

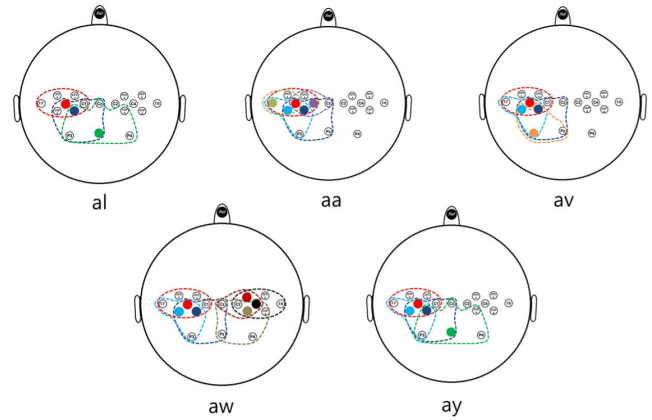


Fig. 8. Channels of selected local regions for LRFCSP. The regions indicated by the dotted line is the selected local regions. The marked channel is the main channel of each selected local region.

TABLE VI

SELECTED FILTER BANKS OF 18 LOCAL REGIONS FOR SUBJECT “aw”

local region	Selected filter banks	local region	Selected filter banks
region 1	8-12Hz, 12-16Hz	region 10	8-12Hz, 12-16Hz
region 2	12-16Hz, 20-24Hz	region 11	12-16Hz, 24-28Hz
region 3	8-12Hz, 12-16Hz	region 12	12-16Hz, 24-28Hz
region 4	12-16Hz, 24-28Hz	region 13	12-16Hz, 24-28Hz
region 5	12-16Hz, 24-28Hz	region 14	12-16Hz, 24-28Hz
region 6	8-12Hz, 12-16Hz	region 15	12-16Hz, 20-24Hz
region 7	8-12Hz, 12-16Hz	region 16	12-16Hz, 20-24Hz
region 8	12-16Hz, 16-20Hz	region 17	12-16Hz, 20-24Hz
region 9	12-16Hz, 20-24Hz	region 18	12-16Hz, 24-28Hz

least training data. Thus, the LRFCSP shows the performance improvement in SSS environment.

In contrast to filter-bank based methods (the FBCSP and FBRCS) that extract the distinct frequency bands for all measurement channels, the LRFCSP extracts distinct frequency bands for local regions. The Table VI shows the two selected filter bank of 18 local regions for subject “aw”. The six local regions included in set Q_{freq} are bolded. As shown in Table VI, the 12-16Hz frequency band is selected for all local regions, but not all local regions select the same frequency band. The selected frequency bands differ even among local regions in the same measurement domain. The six selected local regions employ four frequency bands (8-12Hz, 12-16Hz, 20-24Hz, and 24-28Hz). Therefore, our method eschews the practice of selecting frequency bands to explore entire channels; our LRFCSP improves performance by selecting distinct frequency bands for each local region.

Our proposed method uses two-step discriminative local region selection using VRDS and ICFD. Table VII lists the 18 local regions in VRDS order for subject “ay”; the regions included in set H_{freq} are bolded. Table VII also lists the local regions of set H_{freq} by ICFD order for subject “ay”; the local regions included in the set Q_{freq} are bolded. Table VII shows that the VRDS and ICFD sort orders for local regions differed. Local region 8 of subject “ay” is associated with the second highest VRDS but is not selected by ICFD. Thus, our proposed

TABLE VII
VRDSs AND ICFDs OF LOCAL REGIONS FOR SUBJECT “ay”

local region number	6	8	4	7	5	10	1	3	2	16	14	17	18	11	15	13	12	9
VRDS	1.440	1.437	1.433	1.420	1.312	1.236	1.109	1.046	1.046	0.953	0.764	0.744	0.728	0.695	0.668	0.586	0.572	0.543

(a)

local region number	4	7	6	10	5	2	3	8	1
ICFD	1.673	1.669	1.576	1.510	1.463	1.409	1.397	1.386	1.332

(b)

TABLE VIII
10 × 10 CROSS-VALIDATION CLASSIFICATION ACCURACY OF PROPOSED METHODS AND DIFFERENT METHODS USING BCI COMPETITION III DATASET IVA

subject	CSP-TSM	SBLFB	TFPO-CSP	I-DFBCSP	LRCSP	LRFCSP
al	97.86±3.53	98.64±2.03	98.93±1.73	98.86±1.03	98.93±2.41	98.93±1.73
aa	83.21±6.29	83.21±8.93	86.79±5.05	91.21±5.16	83.57±8.62	93.21±4.89
av	75.1±9.1	75.95±8.29	76.19±7.78	75.95±8.29	80.36±7.76	81.79±6.17
aw	95.47±2.8	96.79±3.13	94.84±2.59	96.79±3.31	90.71±5.11	93.21±5.44
ay	96.79±2.53	95.57±3.50	95.63±3.47	95.57±3.5	96.07±3.12	97.5±1.73
mean	89.69±4.85	90.03±5.18	90.48±4.12	91.68±4.48	89.93±5.4	92.93±3.99

TABLE IX
10 × 10 CROSS-VALIDATION CLASSIFICATION ACCURACY OF PROPOSED METHODS AND DIFFERENT METHODS USING BCI COMPETITION IV DATASET I

subject	CSP-TSM	SBLFB	TFPO-CSP	I-DFBCSP	LRCSP	LRFCSP
a	88.1±6.6	80.9±9.73	87.5±4.68	85.7±9.26	88.4±6.03	87.4±6.71
b	59.1±10.62	58.5±11.12	55±3.26	57±10.74	68.4±6.84	70±5.11
c	67.9±8.7	66.8±12.53	67.5±7.45	69±9.85	72.9±5.09	67.4±6.79
d	84.3±7.9	88.5±7.91	85±8.66	93.4±5.57	85.6±8.36	92.9±6.2
e	90.2±5.8	88.4±6.88	87.5±7.17	91.9±6.92	92.8±4.29	93.4±5.13
f	85.9±7.1	78.8±11.98	85±4.81	86.6±8.48	87.1±5.09	88.8±4.28
g	92.2±5.7	94.1±5.41	92.5±7.2	92.8±5.26	92.6±5.8	93.2±6.21
mean	81.1±7.4	79.4±9.36	80±6.18	82.3±8.01	84±5.93	84.7±5.78

method effectively extract the *good* features using two criteria and improves classification accuracy.

D. Performance Comparison of Proposed Methods With Related Methods Using Cross-Validation

To confirm that the proposed methods are not overfitted to a specific dataset and training data, we test the algorithms for various data sets using cross-validation.

We first compare the cross-validation results of the proposed methods with those of other recent related methods using BCI competition III dataset IVa. Table VIII shows the 10 × 10 cross-validation results of the proposed methods and CSP-TSM, SBLFB, TFPO-CSP, and I-DFBCSP. LRFCSP achieves better classification accuracy than all other methods. CSP-TSM fuses CSP variance features and TSM features, such that a single-frequency band is sufficient to obtain optimal results. [33]. LRCSP uses a wide-frequency band and shows similar performance to CSP-TSM; however, LRFCSP uses subject-related local region with distinct frequency bands, which leads to a mean classification accuracy improvement of 3.24% over CSP-TSM. SBLFB applies sparse Bayesian learning to extract significant features using a linear discriminant criterion [34], and TFPO-CSP uses temporal filter

parameter optimization to improve performance [35]. SBLFB and TFPO-CSP shows higher mean classification accuracy than CSP-TSM and LRCSP (the one without frequency optimization). I-DFBCSP uses discriminative filter bank selection based on mutual information [36]; this algorithm selects four distinct frequency bands considering filter banks with 4-Hz bandwidths and wide-frequency band (7-30Hz). I-DFBCSP achieves a mean classification accuracy of 91.68% higher than CSP-TSM, SBLFB and TFPO-CSP. However, the frequency optimized LRFCSP shows the highest mean classification accuracy.

We also evaluate the performance of the proposed methods using BCI competition IV dataset I and dataset Iib. Table IX shows the 10×10 cross-validation classification accuracy results of CSP-TSM, SBLFB, TFPO-CSP, I-DFBCSP and the proposed methods using BCI competition IV dataset I. LRFCSP also achieves the highest classification accuracy in this evaluation.

BCI competition IV dataset Iib contains only three channels; three local regions can thus be configured using the proposed algorithms. Interestingly, the proposed methods yield the reasonable performance improvement using small channel set. Table X shows the 10×10 cross-validation results of the proposed methods and CSP-TSM, SBLFB, and I-DFBCSP.

TABLE X
10 × 10 CROSS-VALIDATION CLASSIFICATION ACCURACY OF PROPOSED METHODS AND DIFFERENT
METHODS USING BCI COMPETITION IV DATASET IIB

subject	CSP-TSM	SBLFB	I-DFBCSP	LRCSP	LRFCSP
B0103T	77.5±9.44	78.25±9.96	80.75±10.48	75.63±6.88	82.5±7.1
B0203T	60.5±10.43	59.25±11.99	58.37±10.23	55.72±11.62	60±8.94
B0303T	50.94±11.09	49.32±13.34	56±13.06	54.38±11.71	62.78±13.49
B0403T	99.25±2.23	99.12±0.73	99.37±0.60	96.86±4.42	99.37±1.98
B0503T	82.82±10.25	92.04±6.52	90.58±7.96	88.75±5.74	87.5±5.9
B0603T	76.99±9.35	79.49±8.23	82±9.91	65.63±8.96	81.88±10.4
B0703T	86.19±7.89	92.5±6.44	88.87±7.61	83.13±7.25	90.63±7.37
B0803T	88.56±7.95	88.87±8.95	89.5±5.85	82.5±8.74	91.25±6.72
B0903T	80.25±9.47	80.62±10.58	83.75±9.36	85.63±8.86	83.13±10.22
mean	78.11±8.67	79.94±8.73	81.02±8.48	76.47±8.24	82.12±8.01

I-DFBCSP outperforms CSP-TSM and SBLFB. However, LRFCSP, the frequency optimized version, shows the highest mean classification performance compared to other methods.

V. CONCLUSION

We describe a novel feature extraction approach for MI classification, which we termed the LRCSP, and its frequency-optimized version, the LRFCSP. The LRCSP extracts CSP features from the local regions defined for each channel. LRCSP selects *good* local regions using VRDS and ICFD in order to improve classification performance. The LRCSP performs better than does the CSPP, which utilizes limited channel subsets. The LRFCSP extends the LRCSP to accommodate two frequency-band signals from each local region. The two frequency bands are selected to optimize mutual information for each local region. LRFCSP performs substantially better than the existing frequency-optimized CSP-based methods FBCSP, SBRCSPP and FBRCSPP. The LRFCSP enhances MI classification, especially in SSS. Also, LRFCSP outperforms other recent related methods in cross-validation results.

REFERENCES

- [1] B. Blankertz, G. Dornhege, M. Krauledat, K.-R. Müller, and G. Curio, "The non-invasive Berlin brain-computer interface: Fast acquisition of effective performance in untrained subjects," *NeuroImage*, vol. 37, no. 2, pp. 539–550, 2007.
- [2] S. Gao, Y. Wang, X. Gao, and B. Hong, "Visual and auditory brain-computer interfaces," *IEEE Trans. Biomed. Eng.*, vol. 61, no. 5, pp. 1436–1447, May 2014.
- [3] E. A. Curran and M. J. Stokes, "Learning to control brain activity: A review of the production and control of EEG components for driving brain-computer interface (BCI) systems," *Brain Cognit.*, vol. 51, no. 3, pp. 326–336, Apr. 2003.
- [4] G. Pfurtscheller and F. H. L. da Silva, "Event-related EEG/MEG synchronization and desynchronization: Basic principles," *Clin. Neurophysiol.*, vol. 110, no. 11, pp. 1842–1857, Nov. 1999.
- [5] C. Neuper, R. Scherer, S. Wriessnegger, and G. Pfurtscheller, "Motor imagery and action observation: Modulation of sensorimotor brain rhythms during mental control of a brain-computer interface," *Clin. Neurophysiol.*, vol. 120, no. 2, pp. 239–247, Feb. 2009.
- [6] C. Guger, H. Ramoser, and G. Pfurtscheller, "Real-time EEG analysis with subject-specific spatial patterns for a brain-computer interface (BCI)," *IEEE Trans. Rehabil. Eng.*, vol. 8, no. 4, pp. 447–456, Dec. 2000.
- [7] B. Blankertz, R. Tomioka, S. Lemm, M. Kawanabe, and K.-R. Müller, "Optimizing spatial filters for robust EEG single-trial analysis," *IEEE Signal Process. Mag.*, vol. 25, no. 1, pp. 41–56, Dec. 2008.
- [8] H. Ramoser, J. Müller-Gerking, and G. Pfurtscheller, "Optimal spatial filtering of single trial EEG during imagined hand movement," *IEEE Trans. Rehabil. Eng.*, vol. 8, no. 3, pp. 441–446, Dec. 2000.
- [9] J. Müller-Gerking, G. Pfurtscheller, and H. Flyvbjerg, "Designing optimal spatial filters for single-trial EEG classification in a movement task," *Clin. Neurophysiol.*, vol. 110, no. 5, pp. 787–798, May 1999.
- [10] H. Wang, Q. Tang, and W. Zheng, "L1-norm-based common spatial patterns," *IEEE Trans. Biomed. Eng.*, vol. 59, no. 3, pp. 653–662, Mar. 2012.
- [11] J. H. Friedman, "Regularized discriminant analysis," *J. Amer. Statist. Assoc.*, vol. 84, no. 405, pp. 165–175, Mar. 1989.
- [12] H. Lu, K. N. Plataniotis, and A. N. Venetsanopoulos, "Regularized common spatial patterns with generic learning for EEG signal classification," in *Proc. EMBC*, 2009, pp. 6599–6602.
- [13] F. Lotte and C. Guan, "Regularizing common spatial patterns to improve BCI designs: Unified theory and new algorithms," *IEEE Trans. Biomed. Eng.*, vol. 58, no. 2, pp. 355–362, Feb. 2011.
- [14] H. Lu, H.-L. Eng, C. Guan, K. N. Plataniotis, and A. N. Venetsanopoulos, "Regularized common spatial pattern with aggregation for EEG classification in small-sample setting," *IEEE Trans. Biomed. Eng.*, vol. 57, no. 12, pp. 2936–2945, Dec. 2010.
- [15] C. Sannelli, C. Vidaurre, K.-R. Müller, and B. Blankertz, "Common spatial pattern patches: An optimized filter ensemble for adaptive brain-computer interfaces," *J. Neural Eng.*, vol. 8, no. 2, 2011, Art. no. 025012.
- [16] C. Sannelli, C. Vidaurre, K.-R. Müller, and B. Blankertz, "Common spatial pattern patches: Online evaluation on BCI-naïve users," in *Proc. 34th Annu. Int. Conf. IEEE EMBS*, Aug./Sep. 2012, pp. 4744–4747.
- [17] Q. Novi, C. Guan, T. H. Dat, and P. Xue, "Sub-band common spatial pattern (SBCSP) for brain-computer interface," in *Proc. 3rd Int. IEEE/EMBS Conf. Neural Eng.*, May 2007, pp. 204–207.
- [18] Y. Liu, H. Zhang, M. Chen, and L. Zhang, "A boosting-based spatial-spectral model for stroke patients' EEG analysis in rehabilitation training," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 24, no. 1, pp. 169–179, Jan. 2015.
- [19] K. K. Ang, Z. Y. Chin, H. Zhang, and C. Guan, "Filter bank common spatial pattern (FBCSP) in brain-computer interface," in *Proc. IEEE Int. Joint Conf. Neural Netw.*, Jun. 2008, pp. 2390–2397.
- [20] K. K. Ang, Z. Y. Chin, C. Wang, C. Guan, and H. Zhang, "Filter bank common spatial pattern algorithm on BCI competition IV datasets 2a and 2b," *Frontiers Neurosci.*, vol. 6, p. 39, Mar. 2012.
- [21] S.-H. Park and S.-G. Lee, "Small sample setting and frequency band selection problem solving using subband regularized common spatial pattern," *IEEE Sensors J.*, vol. 17, no. 10, pp. 2977–2983, May 2017.
- [22] S.-H. Park, D. Lee, and S.-G. Lee, "Filter bank regularized common spatial pattern ensemble for small sample motor imagery classification," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 26, no. 2, pp. 498–505, Feb. 2018.
- [23] Y. Park and W. Chung, "BCI classification using locally generated CSP features," in *Proc. 6th Int. Conf. Brain-Comput. Interface (BCI)*, Jan. 2018, pp. 1–4.
- [24] J. Holsheimer and B. Feenstra, "Volume conduction and EEG measurements within the brain: A quantitative approach to the influence of electrical spread on the linear relationship of activity measured at different locations," *Electroencephalogr. Clin. Neurophysiol.*, vol. 43, no. 1, pp. 52–58, 1977. [Online]. Available: <http://doc.utwente.nl/68036/>

- [25] N. Rafiuddin, Y. U. Khan, and O. Farooq, "Feature extraction and classification of EEG for automatic seizure detection," in *Proc. Int. Conf. Multimedia, Signal Process. Commun. Technol.*, Aligarh, India, 2011, pp. 184–187.
- [26] R. Battiti, "Using mutual information for selecting features in supervised neural net learning," *IEEE Trans. Neural Netw.*, vol. 5, no. 4, pp. 537–550, Jul. 1994.
- [27] L. Yu and H. Liu, "Feature selection for high-dimensional data: A fast correlation-based filter solution," in *Proc. 20th Int. Conf. Mach. Learn.*, 2003, pp. 856–863.
- [28] (2005). *Data set IVa for the BCI Competition III*. [Online]. Available: http://www.bbc.de/competition/iii/desc_IVa.html
- [29] B. Blankertz *et al.*, "The BCI competition III: Validating alternative approaches to actual BCI problems," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 14, no. 2, pp. 153–159, Jun. 2006.
- [30] Y. Zhang, G. Zhou, J. Jin, X. Wang, and A. Cichocki, "Optimizing spatial patterns with sparse filter bands for motor-imagery based brain-computer interface," *J. Neurosci. Methods*, no. 255, pp. 85–91, Nov. 2015.
- [31] J. Ye and T. Xiong, "SVM versus least squares SVM," in *Proc. 11th Int. Conf. Artif. Intell. Statist. (AISTATS)*, vol. 2, Mar. 2007, pp. 644–651.
- [32] C. C. J. M. de Klerk, M. H. Johnson, and V. Southgate, "An EEG study on the somatotopic organisation of sensorimotor cortex activation during action execution and observation in infancy," *Develop. Cogn. Neurosci.*, vol. 15, pp. 1–10, Oct. 2015.
- [33] S. Kumar, K. Mamun, and A. Sharma, "CSP-TSM: Optimizing the performance of Riemannian tangent space mapping using common spatial pattern for MI-BCI," *Comput. Biol. Med.*, vol. 91, pp. 231–242, Dec. 2017.
- [34] Y. Zhang, Y. Wang, J. Jin, and X. Wang, "Sparse Bayesian learning for obtaining sparsity of EEG frequency bands based feature vectors in motor imagery classification," *Int. J. Neural Syst.*, vol. 27, no. 2, 2017, Art. no. 1650032.
- [35] S. Kumar and A. Sharma, "A new parameter tuning approach for enhanced motor imagery EEG signal classification," *Med. Biol. Eng. Comput.*, vol. 56, no. 4, pp. 1–14, Apr. 2018.
- [36] S. Kumar, A. Sharma, and T. Tsunoda, "An improved discriminative filter bank selection approach for motor imagery EEG signal classification using mutual information," *BMC Bioinform.*, vol. 18, p. 545, Dec. 2017.